# Checkpoint 2 Summary

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## 1. Changes in the Checkpoint

For Checkpoint 2, we built upon the basic functionality implemented in the first checkpoint and focused on improving the system performance and making it capable of handling real-world double-talk. Here is what we changed:

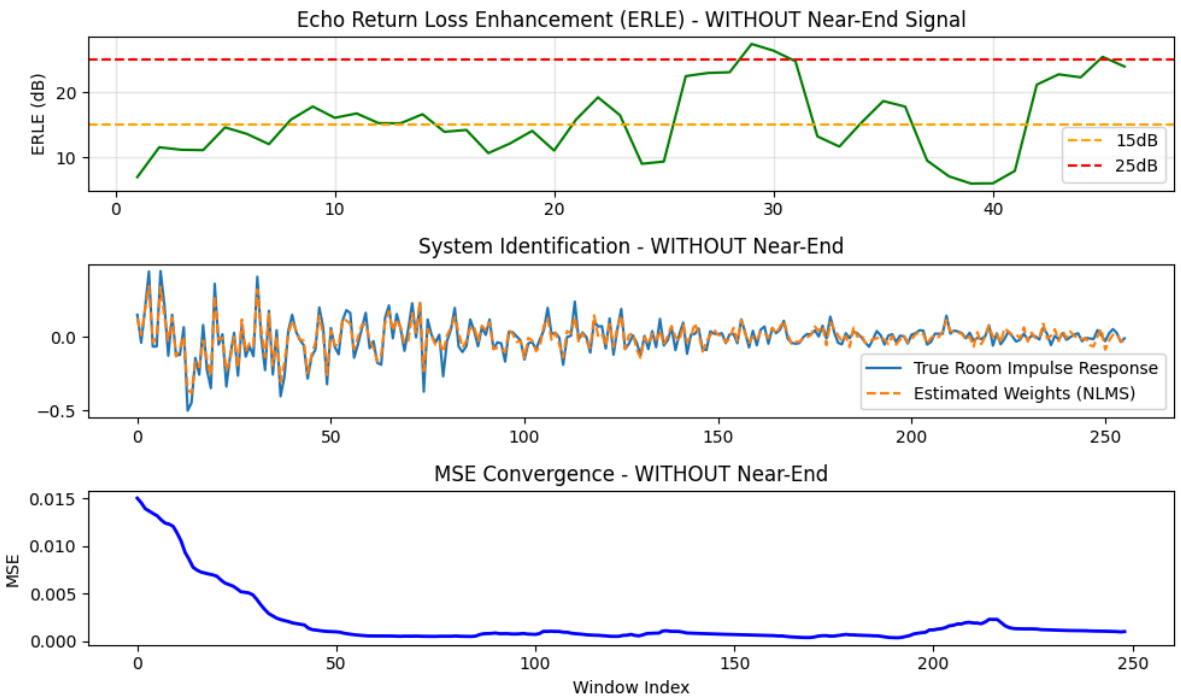
- **Fixed Normalization:** We were normalizing the signals before sending them into the algorithm, which was artificially squashing the echo, hurting the accuracy. Now, we feed raw arrays into the functions and only normalize at the end for audio playback and writing to the .wav files.
- **Added Double-Talk Detection:** We implemented a Geigel-style detector inside the NLMS loop. If the microphone input crosses a certain threshold (indicating the near-end person is speaking loudly), the algorithm skips the weight update step. This freezes the filter and protects our learned weights from being corrupted.
- **Tweaked Hyperparameters:** We tested out various values for the parameters before finally settling on the final ones. `FILTER_SIZE` = 256, `DECAY_CONSTANT` = 0.01, `STEP_SIZE` = 1, and `REGULARIZATION` = 0.0001.
- **Expanded Testing & Metrics:** We expanded our testing to cover both Single-Talk (near-end absent) and Double-Talk (near-end present) scenarios. We also added a Mean Squared Error (MSE) plot to track how fast the filter minimizes the error.

## 2. Results and Observations

With the code changes and parameters tuning, the system's performance improved in both test cases:

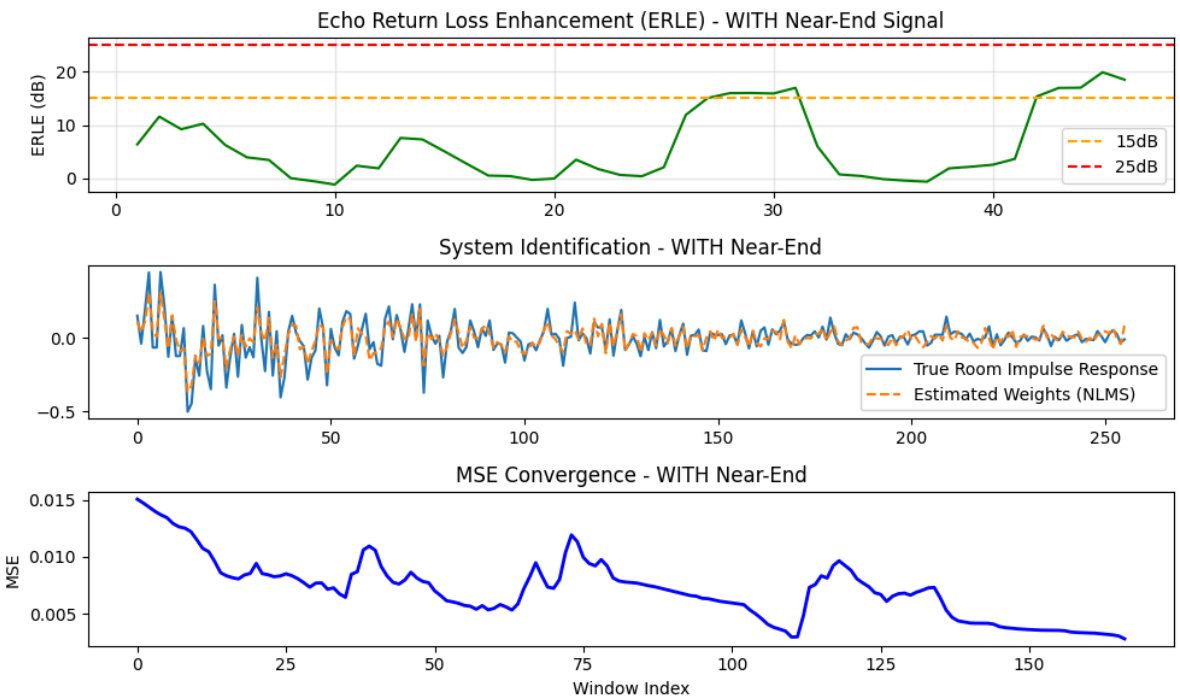
### Case A: WITHOUT Near-End (Single-Talk)

- **Final ERLE:** 23.98 dB
- Observation*:* Without local speech interference, the filter works very well, achieving a high ERLE. The plots show our estimated weights successfully estimating the true room response, and the MSE drops smoothly to near zero.




**Case B: WITH Near-End (Double-Talk)**

- **Final ERLE:** 18.50 dB
- Observation*:* The ERLE naturally drops slightly because the local voice acts as "noise" in the calculation, but the result is still above the 15 db threshold. More importantly, the System Identification plot shows that the weights remain stable instead of diverging. This means that the Geigel detector successfully froze the weights update during overlapping speech.



However, we observed that the audio quality of the cleaned_output.wav is still not perfect. The echoed far-end signal is not fully eliminated, leaving behind some noise.

### 3. Conclusion

The updates in Checkpoint 2 transformed our initial NLMS script into a stable digital signal processing system. Fixing the normalization bug allowed the filter to finally identify the true room acoustics, and the Double-Talk detector prevented the distortion we were previously getting during overlapping speech. The system can now better suppress echo while preserving the near-end audio.